

Conservation decision making: integrating the precautionary principle with uncertainty

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The precautionary principle suggests that more conservative actions are prudent when levels of uncertainty are high. Decision makers often rely on intuition rather than specific methods to implement the precautionary principle. I demonstrate how Bayesian data analysis produces uncertainty metrics that can be easily blended with a mathematically explicit rendition of the precautionary principle. This results in decisions that are transparent, replicable, and exact. Societal values will often determine appropriate levels of precaution, so methods that elicit input from public stakeholders are needed.

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The precautionary principle is captured in aphorisms such as “better safe than sorry” and “an ounce of prevention is worth a pound of cure”. These statements seem to convey good advice even though they do not have literal meaning. For example, it is obvious that safety is better than sorrow; the real issue concerns the magnitude of the reward associated with risky behavior. Informally, the precautionary principle grapples with what seems like a comparison between apples and oranges and concludes that this beleaguered endeavor should be resolved conservatively. While several definitions have been provided, (Johnson 2012), the precautionary principle is perhaps best known in an environmental setting for saddling public policy advocates with the burden of proof that their proposal is harmless (UNCED 1992). This meaning seems to suffer from the same oversimplification as “better safe than sorry” because it does not attempt to blend the utility of potential outcomes with an associated probability of occurrence.

Here, I broadly define the precautionary principle as a *manifestation* of risk aversion and loss aversion. Risk aversion is a measure of disdain for uncertainty. For instance, some people would prefer to receive a guaran-

teed payment of \$1000 instead of entering a lottery with a 50:50 chance of winning either \$2100 or nothing. If the choice were repeated in n iterations, choosing the guaranteed payout yields $\$1000n$ and choosing the lottery yields an average of $(0.5 \times \$2100 + 0.5 \times \$0 =) \$1050n$. Choosing the lottery is therefore rational because it will lead to $\$50n$ more money, on average, than repeatedly choosing the guaranteed payment. A key psychological factor is revealed when repeat iterations of the choice are not possible. Many people prefer the guaranteed payout rather than to face the possibility of getting nothing. If the choice is reframed as a guaranteed payment of \$1,000,000 versus a lottery with a 50:50 chance of winning \$2,000,100 or nothing, then the lottery once again has a $\$50n$ advantage, but more people will now prefer the guaranteed payment. People who change their decision when presented with different payment amounts may appear irrational to others because the mathematical parameters of the decision do not seem to change. However, the benefits that people reap from money have diminishing returns with increasing amounts of money, and so people may indeed make internally coherent decisions based on subjective values about the usefulness of money. This phenomenon was described 278 years ago by the 18th-century Swiss mathematician Bernoulli (Bernoulli 1954). Sometimes, uncertainty cannot be measured or is not known quantitatively. This is sometimes referred to as “ambiguity” (Johnson 2012). While this form of uncertainty also contributes to risk aversion and the precautionary principle, the approaches described below apply to quantified uncertainty about the state of nature, as revealed by statistical analysis.

Loss aversion is a measure of people’s preference for avoiding loss rather than acquiring benefit (Kahneman and Tversky 1984). For example, people may be more motivated to prevent the loss of \$1000 than to acquire \$1000. Furthermore, the magnitude of this extra motivation can be influenced by the magnitude of

In a nutshell:

- Is a population with a small chance of massive decline a greater conservation priority than a population with a high chance of moderate decline?
- Decision making can be fraught with subjectivity and cognitive biases
- Optimal conservation decisions depend on the level of precaution that society supports
- Standardized methods will help resource managers communicate value-laden concepts to public stakeholders and implement complex conservation plans

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the loss/benefit. This concept plays a central role in prospect theory (Kahneman and Tversky 1979), for which Kahneman received a Nobel Prize in Economics. Loss aversion suggests irrationality because a decision maker's preference can be affected by how the decision is described or framed (Kahneman and Tversky 1984). Conservation managers face the additional challenge of addressing outcomes that, unlike loss of money, are not reversible (Gollier *et al.* 2000). The irreversibility of extinction makes it difficult to compare with its complement – persistence. Loss aversion in a conservation context therefore includes a judgment about the value of extinction and persistence. The social context within which different decision makers work may place different costs/rewards for presiding over extinction versus persistence, and therefore

can affect the magnitude of loss aversion experienced by decision makers.

Conservation decision making understandably includes subjectivity associated with risk aversion and loss aversion. It is perhaps more unsettling that the past several decades of psychological research on decision making about uncertain outcomes have revealed a number of cognitive biases (Tversky and Kahneman 1974; Denes-Raj and Epstein 1994; Hilbert 2012). For example, pigeons (*Columba livia*) seem to grasp the optimal solution to a classic brain teaser, the “Monty Hall problem”, faster than people (https://en.wikipedia.org/wiki/Monty_Hall_problem; Hebranson and Schroeder 2010). Indeed, cognitive biases associated with decision making are regularly exploited by astute businesspeople, and help explain economic behavior that cannot

Panel 1. Bayesian uncertainty

The difference between “frequentist” and “Bayesian” paradigms of statistical inference has been the subject of a longstanding controversy that has been well documented in ecological literature (Ellison 2004; Clark 2005; Cressie *et al.* 2009; Lele and Dennis 2009). The frequentists are so named because they define probability as the frequency of a given outcome in a large number of trials. Frequentists use a probability model for the sampling distribution of the data (y) given one or more unknown parameters (θ). This model can be written simply as $P(y | \theta)$ to denote the probability of data given unknown parameters. However, since data are known empirically and parameters require inference, the roles of data and parameters must be reversed. This results in a likelihood (L) of parameters given the data:

$$L(\theta | y) = P(y | \theta) \neq P(\theta | y).$$

Frequentist inference proceeds by finding values for θ that maximize the likelihood (ie the probability of the data). Uncertainty is characterized with a confidence interval that is constructed in a manner that will miss a parameter's true value with a stated frequency (known as the α level) if an infinite number of replicate datasets were available. It is a common misinterpretation to conclude that a confidence interval has a probability ($1-\alpha$) of containing the parameter. But the relationships given above indicate that frequentists cannot use probability to describe parameters. Instead, the likelihood yields a relative support of different values of parameters, and an attempt to convert this to probability by integrating over regions of the parameter space yields results that are sensitive to how the parameters are defined (ie L is not invariant to parameterization; Fisher 1912; Aldrich 1997).

A probabilistic interpretation of parameters is achieved by Bayesians, who define probability as a “degree of belief”. Like frequentists, Bayesians also use $L(\theta | y) = P(y | \theta)$, but they blend it with a probability distribution representing degrees of belief in the parameter prior to inspecting data, $P(\theta)$. The blending follows from undisputed axioms of probability, but the source of this prior information is potentially arbitrary, and so the Bayesian paradigm is criticized by frequentists for permitting subjectivity. Bayesians contend that arbitrariness can be controlled if the proper prior belief probability distributions are supplied (but see Lele and Dennis 2009), or even embrace the

subjectivity as the mathematically correct synthesis of an actual prior belief and new data. Because only the Bayesian paradigm arrives at a probability of the parameter(s) given the data, $P(\theta | y)$, Bayesians are uniquely able to answer the question, “What is the probability that this population has a negative trend?” A dedicated frequentist must argue that the question is flawed because the population is either declining or not declining, and uncertainty in which category the population is categorized is predicated on confidence intervals with a relatively difficult interpretation that is not immediately useful to a risk-averse conservation manager.

The Bayesian “blending” of prior information with a likelihood follows from Thomas Bayes' observations about the relationships between joint, conditional, and marginal probabilities. Specifically, the joint probability of two events is the product of the conditional probability and the marginal probability:

$$P(\theta, y) = P(\theta | y)P(y) \quad (\text{Eq P1})$$

$$P(\theta, y) = P(y | \theta)P(\theta) \quad (\text{Eq P2})$$

Solve Equation P1 for $P(\theta | y)$:

$$P(\theta | y) = P(\theta, y) / P(y).$$

Since $P(\theta, y)$ is not known, substitute Equation P2:

$$P(\theta | y) = P(y | \theta)P(\theta) / P(y)$$

where $P(y)$ is a scaling factor that ensures that $P(\theta | y)$ integrates to one and thus satisfies a necessary requirement of probability. The scaling factor $P(y)$, which is the probability of the data, is equal to the likelihood times the prior integrated over all possible parameter values. The mathematics to obtain $P(y)$ are not possible for all but the simplest cases, so the Bayesian method languished until computers could be made to numerically sample the numerator. These numerical methods, broadly known as Markov chain Monte Carlo methods, were invented by the mathematical physicists who developed atomic weapons in Los Alamos, but their utility was not broadly known to statisticians until 1990 (Robert and Casella 2011).

be justified by rational choice theory. Because bias and irrationality are psychophysical realities of decision making, it is reasonable to inquire how conservation decision makers blend uncertainty with the precautionary principle.

■ Example: population trend

Conservation managers must determine which populations are most in need of protections or other remedial actions. Trend in abundance through time is commonly used for this purpose (NOAA 1990; IUCN 2001). Superficially, assessing population trend is straightforward. Simply plot multiple observations or estimates of population size against time, and then look for an “up” or “down” pattern. However, the scatter in such plots begs an unavoidable statistical question: is any apparent trend really just an artifact of randomness in the data? Statisticians and quantitative ecologists have developed sophisticated tools for addressing this question (Dixon and Pechmann 2005; Humbert *et al.* 2009). Yet uncertainty is never eliminated by such analyses; at best, it is appropriately characterized.

Bayesian data analysis is uniquely able to yield a probability-based interpretation of uncertainty in the magnitude of trend (Panel 1). Bayesian uncertainty in the magnitude of trend is given for two hypothetical populations in Figure 1. The area under the curve to the left of 0 equals the probability that the population is declining (Figure 1, top row). The population depicted in blue has a lower probability of a negative trend than the population shown in red ($0.50 < 0.73$). But because there is more uncertainty in the blue population (the distribution is wider), it has a greater probability of a marked decline than the red population (Figure 1, bottom row; $0.09 > 0.03$). Which population presents a greater conservation concern? More generally, how can potential magnitudes of trend and the associated probabilities become integrated into a single metric or decision criterion?

One solution is to simply average over the uncertainty. Since trend is defined on a continuous scale, there are an infinite number of potential magnitudes for trend. Integral calculus is helpful here. We can compute the “expectation” of trend by weighting each potential value x by the associated probability $P(x)$:

$$E(X) = \int_{-\infty}^{\infty} x * P(x) dx. \quad (\text{Eq 1})$$

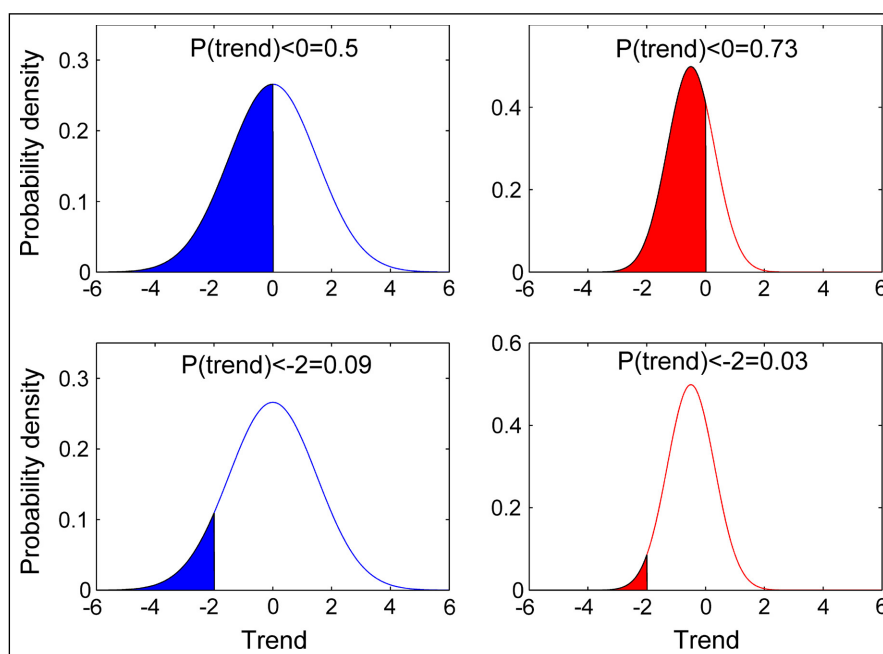


Figure 1. Two hypothetical populations (blue and red; columns) have different Bayesian posterior probability distributions for trend (x axis is percent annual change). Both are normal distributions. The blue population has mean = 0, standard deviation (sd) = 1.5. The red population has mean = -0.5, sd = 0.8. The size of the colored area gives the probability of decline (top row) or the probability of decline of at least -2% per year (bottom row).

However, this does not incorporate the precautionary principle. There is no explicit weighting to capture the fact that many people are more concerned with losses than they are satisfied by gains. For example, the precautionary principle implies that the probability that trend is -2% per year or worse is not perfectly counterbalanced by an identical probability that trend is +2% per year or better. Equation 1 ignores the precautionary principle because it applies weights on trend that depend only on the associated probability of occurrence. Ignoring the precautionary principle is not inherently wrong; it is a judgment. In practice, decision makers may adopt some degree of the precautionary principle when they face uncertainty. This means that decision makers will not average away uncertainty that is symmetrically distributed around the point estimate of trend.

■ The precautionary principle in practice

A simple graphical tool for incorporating the precautionary principle into a trend-based decision begins with the identification of two different critical levels of trend. For instance, choose levels corresponding to any negative trend ($\text{trend} < 0$) and a negative trend of 2% per year or worse ($\text{trend} \leq -2$). Let the probabilities associated with these two levels of trend form the axes of a bivariate plot (Figure 2). Trend categories can be defined by regions within

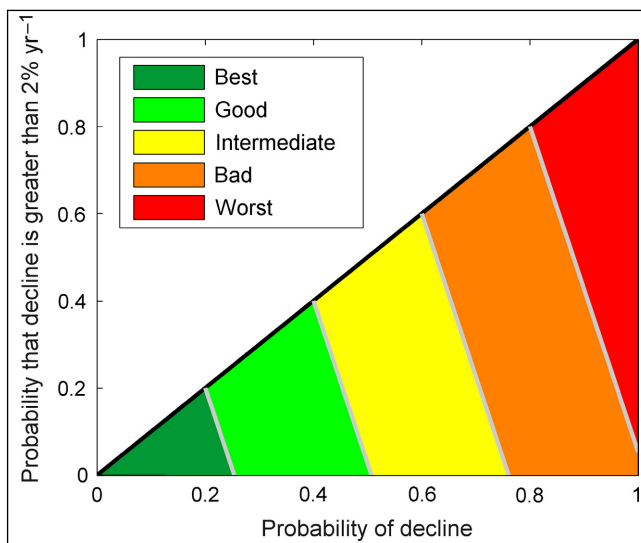


Figure 2. Discretization of posterior probability of trend into five risk categories. The five categories are all below the black diagonal line. The top of the gray lines occur at (0.2, 0.2), (0.4, 0.4), (0.6, 0.6), and (0.8, 0.8), and extend downward at a 60° angle with the black diagonal line. Note that if the gray lines were perfectly vertical (making a 90° angle with the x axis), then risk categories would be independent of the values on the y axis. The angle of the gray line is a “weight” for the importance of the two trend metrics, and reflects loss aversion.

this space, and the position of lines demarcating one category from the next explicitly “weight” the relative importance of the two critical levels of trend.

Another approach to the same problem uses calculus to integrate an aversion function with the uncertainty in trend. Aversion can be modeled with generic units that decay exponentially as trend increases:

$$A = e^{-rx}, \quad (\text{Eq 2})$$

where r describes the rate of decline in aversion (A) over various values of trend x , and where e is a constant (2.718...). It is now possible to compute an aversion-and-uncertainty-weighted quality score:

$$\text{Quality} = \int_{-\infty}^{\infty} x * P(x) * A(x) dx. \quad (\text{Eq 3})$$

For the normal distributions used in Figure 1, Equation 3 can be written more explicitly:

$$\text{Quality} = \int_{-\infty}^{\infty} x \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{-rx} dx.$$

To specify no aversion, set $r = 0$ and Equation 2 will evaluate to 1 for any value of trend, x . This constant then drops out from the integral in Equation 3, making it equivalent to Equation 1. Indeed, the integral of

Equation 3 is simply the mean of the probability distribution for trend when $r = 0$. However, as r increases, there is more aversion to population decline (Figure 3a) and area “accumulates” in the negative regions of the *Quality* function (Equation 3, Figure 3b). The integral then evaluates to a number less than the mean of the probability distribution for trend. Returning to the hypothetical populations presented in Figure 1, Equation 3 indicates that the red population is a greater concern (lower *Quality*) than the blue population when $r = 0$ (Equation 3 = -0.5 and 0 for red and blue, respectively). Yet once r reaches 0.35 , the *Quality* for the red population exceeds that of the blue population (Equation 3 = -0.904 and -0.897 for blue and red, respectively), indicating that the blue population is now the greater conservation concern.

Managing a natural resource

The precautionary principle, as defined here, represents the decision maker’s aversion to uncertainty and the associated fear that any decision will be implicated in the extinction of the target population. In most management scenarios, an analyst will present a statistical trend assessment with some form of uncertainty to a decision maker. The decision maker’s blending of statistical uncertainty with the precautionary principle is a private mental approximation commonly referred to as “gut feeling”. Making an explicit (ie mathematical) statement about the precautionary principle enhances transparency and ensures that difficult probabilistic concepts are properly synthesized.

The precautionary principle is often implemented by management officials for the sake of expedience, but high-stakes decisions should involve public stakeholders (Reed 2008). Getting stakeholders to agree on a mathematical formulation of the precautionary principle represents a key barrier to natural resource decision making. Although methods exist for eliciting quantitative information from people’s minds (Lele and Allen 2006; Aspinall 2010), natural resource stakeholders are not necessarily accustomed to representing ideas quantitatively, and may provide responses they believe will yield a desired outcome. Still, not engaging stakeholders about risk/loss aversion may pose an even greater problem to resource managers. What is needed, then, are tools, examples, and procedures that managers can use with their stakeholders to more effectively elicit levels of precaution and to communicate the decision-making process.

Using Bayesian inference to facilitate environmental decision making is not new (Wolfson *et al.* 1996; Aanesen *et al.* 2014). Indeed, adaptive resource management is often cast in a Bayesian framework because new information can be used to “update” beliefs in competing management alternatives (Conroy and Peterson 2013). Despite the relatively large literature expounding

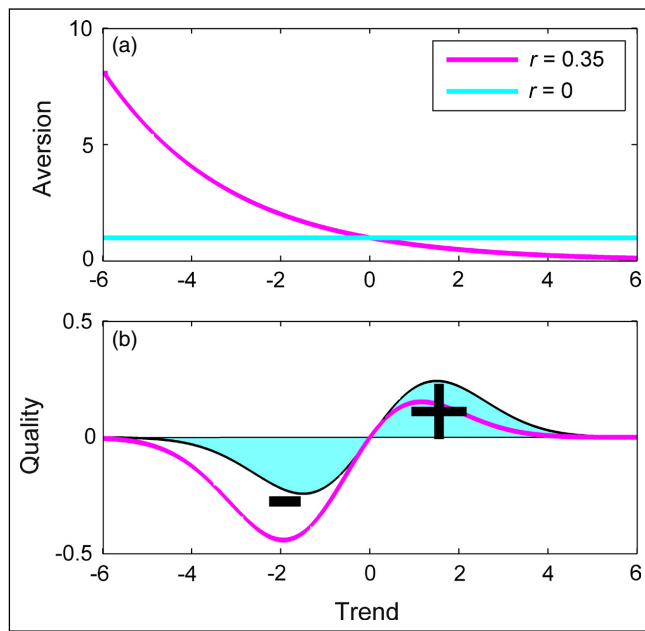


Figure 3. Aversion for various values of trend can be modeled with an exponential function (Equation 2; [a]). When the rate of exponential decline is zero ($r = 0$, cyan), there is no aversion, and the quality function given in Equation 3 will have negative and positive regions that integrate to the mean of the probability distribution for trend (shaded area in [b]). As aversion becomes more severe ($r = 0.35$, magenta), area accumulates in the negative region of the quality function ([b], not shaded), and the integral becomes more negative, indicating lower quality and greater conservation concern.

its virtues, adaptive resource management is rarely used in actual decision making (Westgate *et al.* 2013), suggesting that decision makers need more help implementing this approach. Although the prioritization problem presented here is not adaptive because it does not intend to reduce management uncertainty over time, it is nonetheless a real conservation dilemma that benefited from the use of Bayesian measures of uncertainty in trend (ODFW 2014). In ODFW (2014), trend was assessed for multiple, geographically distributed populations. The people living in the different areas affected by conservation decisions should be concerned about fair and equitable treatment. A mathematically structured conservation decision provides the transparency and exactness needed to resolve the concerns that could be amplified in the case of a more whimsical decision-making process.

A non-Bayesian treatment of the same issue might choose the lower confidence interval of a trend estimate as a hedge against uncertainty. This could also be done with a Bayesian assessment, but the probabilistic interpretation of parameters offered by Bayes (Panel 1) admits a continuous scale of trends useful for integrating the precautionary principle. While it is easier to deal with binary outcomes like “better safe than sorry”, the

difficult reality that conservation managers face concerns a spectrum of costs/benefits associated with uncertain outcomes. The methods discussed here are designed to address this.

Trend is a conservation metric that can be computed for any species in any ecosystem. This universality provides an opportunity for more standardized methods. Like the geneticists’ “model organism” (eg fruit fly *Drosophila melanogaster*) that is used to understand basic mechanisms with implications for other species, the integration of uncertainty in trend with the precautionary principle could be viewed as a management “test case”. By focusing on how to communicate statistical uncertainty in trend and blend it with risk aversion and loss aversion, we may be able to demonstrate the possibility and usefulness of quantification in other areas of natural resource decision making.

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